

SOFTWARE REQUIREMENTS SPECIFICATION

Collaborative Study Material & Quiz Platform

Submission 2 · Group 1 · Team 1 · Version 1.1

Project Title	Collaborative Study Material & Quiz Platform
Group Number	1
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Methodology	Agile (iterative, sprint-based)
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1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) defines the functional, non-functional and security requirements for the **Collaborative Study Material & Quiz Platform**, version 1.1. It is the authoritative reference for design, implementation, testing and validation of the system and forms the basis of the Requirement Traceability Matrix.

1.2 Document Conventions

- Functional requirements are tagged `FR-xxx`, non-functional `NFR-xxx`, security `SEC-xxx`.
- Each requirement has a **priority** (**High** / **Medium** / **Low**) and a **verification method** (Test / Demonstration / Inspection / Analysis).
- “The system” refers to the Collaborative Study Material & Quiz Platform.

1.3 Intended Audience

Course faculty and evaluators, the project team (developers/testers), and future maintainers. Evaluators should read Sections 5–6 and Appendix C; developers should read the full document.

1.4 Product Scope

The platform is a web application that lets students search, browse and attempt topic-linked quizzes with automatic scoring and track their learning progress through analytics, while students with contributor privileges and educators can additionally upload, organize and collaboratively curate study material and create quizzes. It supports the objective of keeping study material and self-assessment in one connected, role-secured place. Out of scope: real-money payments, live streaming, AI content generation, subjective auto-grading, native mobile apps and institution-wide ERP integration.

1.5 References

- SE Level 3 Mini-Project Guidelines (UE24CS341A).
- Submission 1 — Synopsis / Project Proposal, Group 1.
- IEEE Std 830-style SRS structure (Jackfruit Phase-1 template).

2. Overall Description

2.1 Product Perspective

The system is a new, self-contained MERN-stack web product (MongoDB, Express, React, Node.js). It follows a three-tier architecture: a React single-page client, a Node.js/Express REST API, and a MongoDB data store. It has no dependency on legacy systems; external services are limited to email (for password reset) and object/file storage for uploaded material.

2.2 Product Functions

- User registration, authentication and role-based profiles.
- Study material upload, tagging, categorization, search and sharing.
- Collaborative curation (ratings, comments, revision history) and content moderation.
- Quiz creation and management with multiple question types.
- Quiz attempt with automatic scoring and feedback.
- Progress analytics dashboard and leaderboard.

2.3 User Classes and Characteristics

The system defines three stored account roles — `student`, `educator` and `admin`. **Contributor** is *not* a separate account role: it is an additional privilege that an administrator grants to a student, producing the “Student with Contributor Privileges” user class below.

User Class	Stored Role	Characteristics & Usage
Student	<code>student</code>	Primary user. Browses/searches material, attempts quizzes, views own progress and the leaderboard. Non-technical; peak usage before exams.
Student with Contributor Privileges	<code>student</code> + contributor privilege	A student granted the contributor privilege. Additionally uploads and curates material and creates quizzes; helps maintain content quality. Not a distinct database role.
Educator	<code>educator</code>	Trusted author of authoritative material and assessments; may manage own published content.
Administrator	<code>admin</code>	Manages users and role/privilege assignment, and moderates reported content; maintains platform health.

2.4 Operating Environment

- **Client:** Modern desktop/mobile browsers — Chrome, Firefox, Edge, Safari (latest two major versions); responsive web (no native app).
- **Server:** Node.js runtime on Linux; containerized with Docker.
- **Database:** MongoDB.
- **Network:** HTTPS over the public internet.

2.5 Design and Implementation Constraints

- Must be implemented on the MERN stack (MongoDB, Express, React, Node.js) as the primary technology stack.
- Source in Git/GitHub; CI/CD via Jenkins; containerized with Docker.
- Testing with Jest (JavaScript unit/integration tests); static analysis with SonarQube; planning in Jira/GitHub Projects.
- REST API with JSON payloads. Authentication uses short-lived **access tokens (JWT)** together with longer-lived **refresh tokens**; refresh tokens are server-tracked so they can be revoked on logout.

2.6 Assumptions and Dependencies

- Users have internet access and a supported browser.
- An email service is available for password reset.
- File/object storage is available for uploaded study material.
- Team has access to the required toolchain (GitHub, Jenkins, Docker, SonarQube).

3. External Interface Requirements

3.1 User Interfaces

- Responsive web UI with pages for: login/register, material library (list/search), material detail, quiz builder, quiz attempt, results/analytics dashboard, and profile.
- Consistent navigation bar with role-appropriate actions; keyboard-accessible forms with inline validation messages.

3.2 Hardware Interfaces

No special hardware. Any device with a supported browser and internet connection; server runs on standard cloud/VM hardware.

3.3 Software Interfaces

Interface	Type	Purpose
MongoDB	Database driver	Persist users, material, quizzes, attempts
Email service (SMTP/API)	External API	Password-reset emails
File/object storage	External API / filesystem	Store uploaded study material files
REST API	Internal HTTP/JSON	Client-server communication

3.4 Communications Interfaces

All client-server traffic uses HTTPS (TLS). API uses JSON over HTTP; authenticated requests carry a bearer access token (JWT). Email uses SMTP/HTTPS provider API.

4. Analysis Models

4.1 Use Cases

UC ID	Use Case	Primary Actor	Brief
UC-01	Register / Log in	Student, Educator	Create an account and authenticate to access the platform
UC-02	Manage profile	Authenticated user	View and edit personal profile information (role/privilege changes are performed by the Administrator via UC-10)
UC-03	Upload study material	Student w/ Contributor Privileges, Educator	Upload a file with title, description, category, topic and tags
UC-04	Search & browse material	Student, Student w/ Contributor Privileges, Educator	Search/filter material by keyword, tag or category
UC-05	Curate & engage with material	Student, Student w/ Contributor Privileges, Educator, Administrator	Any authenticated user can rate and comment on accessible material; contributors and educators additionally organize (edit the metadata of — title, description, category, topic, tags per FR-018) and may share material they own
UC-06	Create quiz	Student w/ Contributor Privileges, Educator	Build a quiz with questions, options and answer keys
UC-07	Attempt quiz	Student	Answer quiz questions and submit
UC-08	View results & analytics	Student	See score, feedback, history and progress dashboard
UC-09	View leaderboard	Student	Compare scores across a topic/quiz
UC-10	Administer users/content	Administrator	Manage users, assign roles/privileges, and review reported content (dismiss or hide)

UC ID	Use Case	Primary Actor	Brief
UC-11	Report study material	Student, Student with Contributor Privileges, Educator	Flag material with a reason for administrator review

4.2 Use Case Diagram

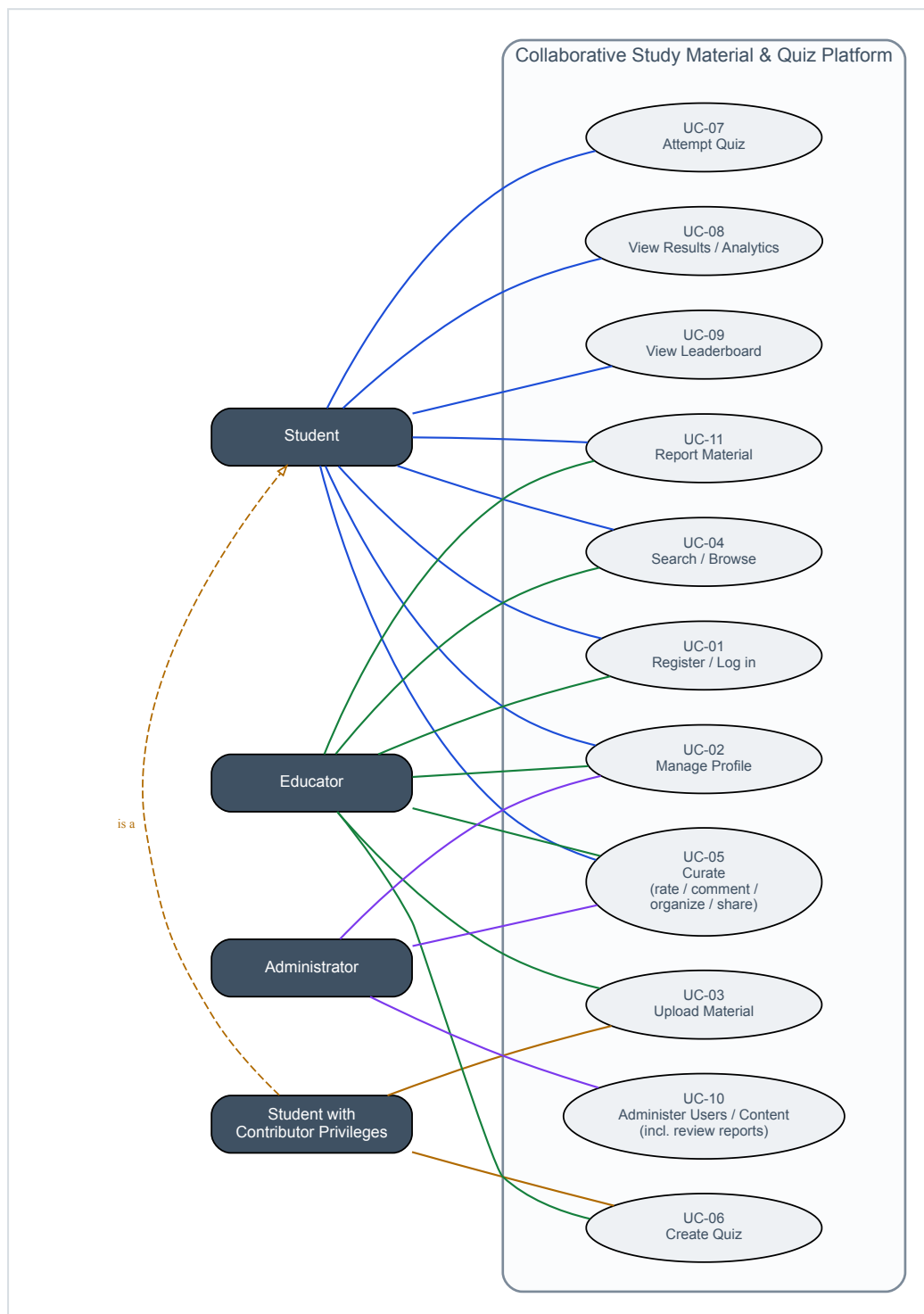


Figure 1 — Use case diagram. “Student with Contributor Privileges” is a generalization of Student (inherits all Student use cases and adds upload/create-quiz).

4.3 Use Case Flow (UC-07 Attempt Quiz — example)

Pre-condition: User is authenticated and a published quiz exists. **Main flow:** (1) Student opens a quiz; (2) system presents questions; (3) student answers and submits; (4) system auto-scores and stores the attempt; (5) system shows score and per-question feedback. **Alternate:** if the student abandons, no attempt is recorded. **Post-condition:** attempt saved and reflected in analytics/leaderboard.

4.4 Conceptual Data Model (ER)

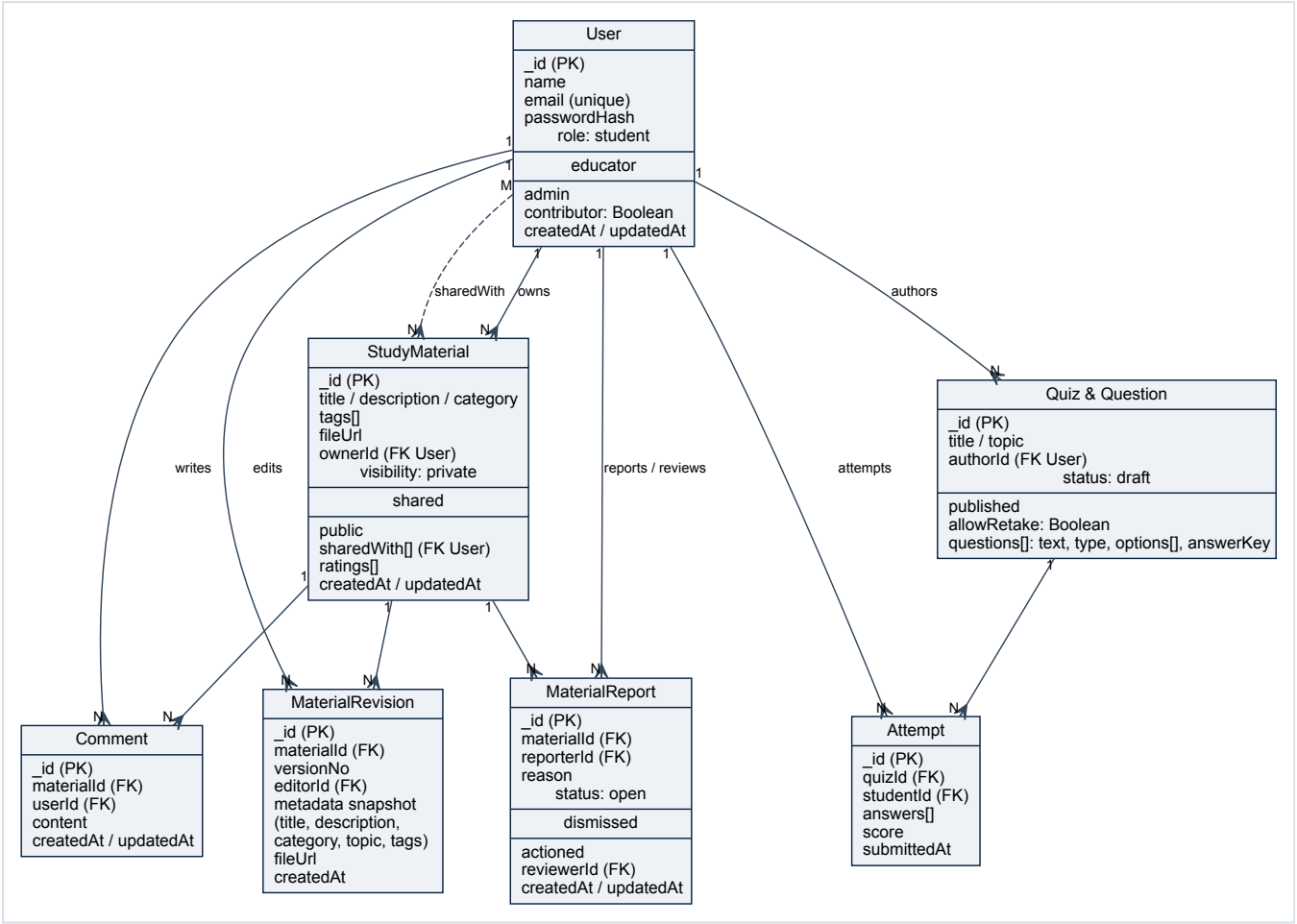


Figure 2 — Conceptual data model. Leaderboard data is derived from stored Attempt scores and is not a separate entity. Field-level detail is in Appendix B.

4.5 Stimulus / Response Sequences

Feature	Stimulus → Response
Authentication	Valid registration input → validate → create account → confirmation. Duplicate/invalid email → reject → inline error. Valid login → issue access + refresh tokens. Logout → revoke refresh token.
Material upload	Authorized user submits file + title/category/tags → validate type/size → store file & metadata → confirmation. Disallowed type/size → reject → error.
Material update	Owner or administrator edits material (file and/or metadata) → save new version → create MaterialRevision capturing the prior state → confirmation.
Quiz creation / publishing	Author defines questions + answer keys → validate (MCQ ≥ 2 options + correct answer; T/F answer) → save as draft. Publish request with ≥ 1 valid question → state draft → published. Publish with no valid question → reject → error.
Quiz attempt	Student submits answers → auto-score against answer key → store attempt → show score + per-question feedback → update analytics & leaderboard (highest score used).
Report / moderation	User reports material with a reason → create MaterialReport (status open) → material temporarily hidden pending review. Administrator reviews → dismiss (material restored) or uphold (material remains hidden).

5. System Features (Functional Requirements)

Requirement IDs are grouped by feature area (FR-001–006 accounts, FR-010–018 study material, FR-020–025 quizzes, FR-030–034 Quiz Attempts, Scoring & Analytics; FR-035 Leaderboard). Gaps between number ranges are intentional grouping, not missing requirements.

5.1 SF-1 User Accounts & Profiles (Owner: Anisha)

ID	Requirement	Priority	Verification
FR-001	The system shall allow a new user to register with name, unique email and password.	High	Test
FR-002	The system shall authenticate a registered user via email and password and issue a short-lived access token (JWT) together with a refresh token.	High	Test
FR-003	The system shall allow an authenticated user to log out, invalidating the active refresh token so it can no longer be used to obtain new access tokens.	Medium	Test
FR-004	The system shall assign each account a system role of student, educator or administrator and enforce role-based access to protected features. The administrator may additionally grant a student the contributor privilege (this privilege is not a separate database role).	High	Test
FR-005	The system shall allow a user to view and edit their own profile details (excluding their role/privilege, which only an administrator can change).	Medium	Demonstration
FR-006	The system shall allow a user to reset a forgotten password via a verified email link.	Medium	Test

5.2 SF-2 Study Material Management & Moderation (Owner: Aryaroop)

ID	Requirement	Priority	Verification
FR-010	The system shall allow a Student with Contributor Privileges or an Educator to upload a study-material file with a title, description, category and topic.	High	Test
FR-011	The system shall allow the uploader to add one or more tags to study material.	Medium	Test
FR-012	The system shall allow users to search and filter material by keyword, tag or category.	High	Test
FR-013	The system shall let an owner set material visibility as private (owner only), shared (owner plus selected users listed in <code>sharedWith</code>), or public (all authorized platform users).	Medium	Demonstration
FR-014	The system shall allow any authenticated platform user to rate and comment on study material they can access, with each comment stored as a separate record.	Medium	Test
FR-015	The system shall create and retain a revision record whenever the material owner or an administrator updates study material (its file and/or its metadata), capturing the previous state of the material so that authorized users can view earlier versions.	Low	Test / Inspection
FR-016	The system shall allow an authenticated Student, Student with Contributor Privileges, or Educator to report study material by selecting a report reason and submitting the report for administrator review.	Medium	Test
FR-017	The system shall allow an administrator to review reported study material and either dismiss the report — restoring the material to general visibility — or uphold it, keeping the material hidden from general users.	Medium	Test
FR-018	The system shall allow the material owner (or an administrator) to edit the metadata of study material they own — its title, description, category, topic and tags; each such edit creates a revision record (FR-015). This is the “organize material” capability referenced by UC-05.	Medium	Test

5.3 SF-3 Quiz Creation, Management & Gamification (Owner: Ayush)

ID	Requirement	Priority	Verification
FR-020	The system shall allow a Student with Contributor Privileges or an Educator to create a quiz with a title and associated topic.	High	Test
FR-021	The system shall support multiple question types (at minimum multiple-choice and true/false).	High	Test
FR-022	The system shall allow the author to define the correct answer key for each question, requiring each multiple-choice question to have at least two options and one defined correct answer, and each true/false question to have a true or false answer.	High	Test
FR-023	The system shall allow the author to edit, publish and unpublish a quiz, and to configure whether re-attempts are allowed (allowRetake).	Medium	Demonstration
FR-024	The system shall allow a quiz to be associated with study material through a shared topic value (topic-based linkage; no direct study-material foreign key is required).	Low	Demonstration
FR-025	The system shall prevent a quiz from being published unless it contains at least one valid question with a defined answer key.	High	Test
FR-035	The system shall display a leaderboard ranking students for a given quiz or topic, ordered by highest score first, with students having equal scores receiving the same rank; a student's highest score is used for ranking. (Consumes attempt scores produced by SF-4.)	Low	Demonstration

Dependency: FR-035 (Ayush) consumes the scores produced by FR-031–FR-033 (Bharat); the leaderboard reads stored Attempt records and does not itself compute scores.

Actor scope (UC-07 / FR-030): attempting quizzes is intentionally a student-facing self-assessment feature. Students — including students with contributor privileges — may attempt quizzes; educators and administrators author, curate and moderate content rather than attempt quizzes.

5.4 SF-4 Quiz Attempts, Scoring & Analytics (Owner: Bharat)

ID	Requirement	Priority	Verification
FR-030	The system shall allow an authenticated student to attempt a published quiz and submit answers, subject to the quiz's re-attempt setting.	High	Test
FR-031	The system shall automatically score a submitted attempt against the answer key, awarding 1 point for each correct answer and 0 for each incorrect answer; a student may submit a quiz with questions left unanswered, and each unanswered question is scored as 0.	High	Test
FR-032	The system shall display the score and per-question feedback immediately after submission.	High	Test
FR-033	The system shall store each attempt and let the student view their attempt history.	Medium	Test
FR-034	The system shall present a progress dashboard summarizing scores over time by topic, using the student's highest score per quiz for progress summaries.	Medium	Demonstration

6. Non-Functional Requirements

6.1 Quality Attributes

ID	Attribute	Measurable Requirement	Verification
NFR-001	Performance	Under a workload of up to 100 concurrent active users (see NFR-002), 95% of page loads shall complete within 3 seconds and quiz auto-scoring shall return within 2 seconds.	Test
NFR-002	Scalability	The system shall support at least 100 concurrent active users without functional error.	Test
NFR-003	Availability	The system shall remain continuously available throughout the scheduled project demonstration/evaluation session (the “evaluation window”), targeting $\geq 99\%$ availability during that window and excluding planned maintenance.	Analysis
NFR-004	Usability	At least 4 of 5 first-time test users shall complete a quiz attempt without external assistance; forms shall show inline validation.	Demonstration
NFR-005	Maintainability	Code shall pass the SonarQube quality gate with no blocker/critical issues and unit-test coverage $\geq 70\%$ on core modules.	Inspection
NFR-006	Portability	The UI shall function correctly on the latest two versions of Chrome, Firefox, Edge and Safari.	Test
NFR-007	Reliability	Submitted quiz attempts shall be persisted with no data loss on normal operation.	Test
NFR-008	Testability	Core business services shall be independently testable, and automated unit tests shall execute without requiring the production UI.	Test

6.2 Safety Requirements

Not applicable. The proposed system is a software-only educational platform and does not directly control physical devices or safety-critical systems.

6.3 Security Requirements

ID	Requirement	Verification
SEC-001	The system shall store passwords only as salted hashes (e.g. bcrypt); plaintext passwords shall never be stored or logged.	Inspection
SEC-002	The system shall authenticate API requests with signed, expiring access tokens (JWT), and shall revoke the refresh token on logout so new access tokens can no longer be issued.	Test
SEC-003	The system shall enforce role-based authorization so users can only access actions permitted for their role/privilege.	Test
SEC-004	The system shall validate and sanitize all inputs to prevent injection and cross-site scripting attacks.	Test
SEC-005	All client–server communication shall use HTTPS/TLS.	Inspection
SEC-006	The system shall validate uploaded files by type and size — permitting only PDF, DOCX, PPTX, PNG and JPG files up to a maximum of 20 MB — and reject any other content.	Test
SEC-007	The system shall enforce resource-ownership authorization so a user cannot modify or delete another user’s material or quizzes unless explicitly authorized or acting as an administrator.	Test
SEC-008	The system shall not expose quiz answer keys to a student before that student submits the attempt.	Test

7. Other Requirements

- **Business / Domain Rules:** BR-1 — only the author or an administrator may edit or unpublish a quiz. BR-2 — a student may attempt a quiz multiple times when the quiz’s `allowRetake` setting permits; the student’s highest score is used for leaderboard ranking and progress summaries. BR-3 — only published quizzes are attemptable. BR-4 — reported material is temporarily hidden pending administrator review; if the report is dismissed the material is restored to general visibility, and if the report is upheld the material remains hidden (see FR-016/FR-017).
- **Data retention:** user data and attempts are retained for the duration of the project/course.
- **Legal / Licensing:** uploaded material must not violate copyright; uploaders confirm they have rights to share.

Appendix A — Glossary

Term	Definition
Study Material	A file/resource (notes, slides, documents) uploaded for study.
Topic	A named academic category/subject label attached to study material and quizzes; the shared key used for material search, quiz linkage, analytics grouping and leaderboard scope. (Represented as a string, not a separate entity in v1.1.)
Curation	Collaborative engagement with study material through rating and commenting by any authenticated user, with additional organization and sharing capabilities available to contributors and educators to improve its quality and discoverability.
Moderation	Administrator review of reported material (dismiss or hide); distinct from curation, which is quality improvement by contributors.
Contributor	A privilege granted to a student (by an administrator) allowing upload, curation and quiz creation. Not a separate database role.
Quiz	A set of questions with defined answer keys, used for self-assessment.
Attempt	A single instance of a student answering and submitting a quiz.
Answer Key	The correct answer(s) defined for a question.
RBAC	Role-Based Access Control — permissions granted by user role/privilege.
Access Token	A short-lived signed JWT presented with each API request to prove authentication.
Refresh Token	A longer-lived, server-tracked token used to obtain new access tokens; revoked on logout.
Revision	A stored snapshot of a prior version of study material — its metadata (title, description, category, topic, tags) and file reference — created whenever the material is updated (FR-015).
MERN	MongoDB, Express, React, Node.js technology stack.
Leaderboard	Ranked list of students by score for a quiz/topic, derived from stored attempt scores.

Appendix B — Database Field Layouts

B.1 User

Field	Type	Notes
_id	ObjectId	Primary key
name	String	Required
email	String	Required, unique
passwordHash	String	Salted hash (SEC-001)
role	Enum	student / educator / admin
contributor	Boolean	Contributor privilege for a student (FR-004)
createdAt / updatedAt	Date	Timestamps

B.2 StudyMaterial

Field	Type	Notes
_id	ObjectId	Primary key
title	String	Required
description	String	Optional
category	String	Required
topic	String	Named topic (see Glossary)
tags	Array<String>	Search index
fileUrl	String	Storage reference
ownerId	ObjectId	FK → User
visibility	Enum	private / shared / public
sharedWith	Array<ObjectId>	FK → User; permitted users when visibility = shared (FR-013)
ratings	Array	userId, value
hidden	Boolean	Hidden pending/after moderation (BR-4, FR-017)
createdAt / updatedAt	Date	Timestamps

B.3 Comment

Field	Type	Notes
_id	ObjectId	Primary key
materialId	ObjectId	FK → StudyMaterial
userId	ObjectId	FK → User (author of comment)
content	String	Comment text (FR-014)
createdAt / updatedAt	Date	Timestamps

B.4 MaterialRevision

Field	Type	Notes
_id	ObjectId	Revision ID (primary key)
materialId	ObjectId	FK → StudyMaterial
versionNo	Number	Incrementing version number
editorId	ObjectId	FK → User (who made the edit)
title	String	Snapshot of the title at this version
description	String	Snapshot of the description at this version
category	String	Snapshot of the category at this version
topic	String	Snapshot of the topic at this version
tags	[String]	Snapshot of the tags at this version
fileUrl	String	Storage reference for the file at this version (unchanged if only metadata was edited)
createdAt	Date	Timestamp (FR-015)

B.5 MaterialReport

Field	Type	Notes
_id	ObjectId	Primary key
materialId	ObjectId	FK → StudyMaterial
reporterId	ObjectId	FK → User (who reported)
reason	String	Report reason (FR-016)
status	Enum	open / dismissed / actioned
reviewerId	ObjectId	FK → User (admin who reviewed; FR-017)
createdAt / updatedAt	Date	Timestamps

B.6 Quiz & Question

Field	Type	Notes
_id	ObjectId	Primary key
title	String	Required
topic	String	Named topic (see Glossary)
authorId	ObjectId	FK → User
status	Enum	draft / published (FR-025 gate)
allowRetake	Boolean	Whether re-attempts are permitted (BR-2)
questions	Array	text, type, options[], answerKey
createdAt / updatedAt	Date	Timestamps

B.7 Attempt

Field	Type	Notes
_id	ObjectId	Primary key
quizId	ObjectId	FK → Quiz
studentId	ObjectId	FK → User
answers	Array	questionId, response
score	Number	Auto-computed (FR-031)
submittedAt	Date	Timestamp

Note: Leaderboard and progress data are derived by querying stored Attempt scores (highest score per student per quiz/topic); no separate Leaderboard collection is required.

Appendix C — Requirement Traceability Matrix

One row per requirement. Architecture / Design / Code columns are **TBD** at the SRS stage and will be completed as the project progresses through design, implementation and testing.

Req	Use Case	Owner	Arch	Design	Code	Verification
FR-001	UC-01	Anisha	TBD	TBD	TBD	Test
FR-002	UC-01	Anisha	TBD	TBD	TBD	Test
FR-003	UC-01	Anisha	TBD	TBD	TBD	Test
FR-004	UC-01, UC-10	Anisha	TBD	TBD	TBD	Test
FR-005	UC-02	Anisha	TBD	TBD	TBD	Demonstration
FR-006	UC-01	Anisha	TBD	TBD	TBD	Test
FR-010	UC-03	Aryaroop	TBD	TBD	TBD	Test
FR-011	UC-03	Aryaroop	TBD	TBD	TBD	Test
FR-012	UC-04	Aryaroop	TBD	TBD	TBD	Test
FR-013	UC-05	Aryaroop	TBD	TBD	TBD	Demonstration
FR-014	UC-05	Aryaroop	TBD	TBD	TBD	Test
FR-015	UC-03	Aryaroop	TBD	TBD	TBD	Test / Inspection
FR-016	UC-11	Aryaroop	TBD	TBD	TBD	Test
FR-017	UC-10	Aryaroop	TBD	TBD	TBD	Test
FR-018	UC-05	Aryaroop	TBD	TBD	TBD	Test
FR-020	UC-06	Ayush	TBD	TBD	TBD	Test
FR-021	UC-06	Ayush	TBD	TBD	TBD	Test
FR-022	UC-06	Ayush	TBD	TBD	TBD	Test
FR-023	UC-06	Ayush	TBD	TBD	TBD	Demonstration
FR-024	UC-06	Ayush	TBD	TBD	TBD	Demonstration
FR-025	UC-06	Ayush	TBD	TBD	TBD	Test
FR-035	UC-09	Ayush	TBD	TBD	TBD	Demonstration
FR-030	UC-07	Bharat	TBD	TBD	TBD	Test
FR-031	UC-07	Bharat	TBD	TBD	TBD	Test
FR-032	UC-07, UC-08	Bharat	TBD	TBD	TBD	Test

Req	Use Case	Owner	Arch	Design	Code	Verification
FR-033	UC-08	Bharat	TBD	TBD	TBD	Test
FR-034	UC-08	Bharat	TBD	TBD	TBD	Demonstration
NFR-001	All	All	TBD	TBD	TBD	Test
NFR-002	All	All	TBD	TBD	TBD	Test
NFR-003	All	All	TBD	TBD	TBD	Analysis
NFR-004	UC-07	All	TBD	TBD	TBD	Demonstration
NFR-005	All	All	TBD	TBD	TBD	Inspection
NFR-006	All	All	TBD	TBD	TBD	Test
NFR-007	UC-07	Bharat	TBD	TBD	TBD	Test
NFR-008	All	All	TBD	TBD	TBD	Test
SEC-001	UC-01	Anisha	TBD	TBD	TBD	Inspection
SEC-002	UC-01	Anisha	TBD	TBD	TBD	Test
SEC-003	All protected use cases (UC-01, UC-03, UC-05, UC-06, UC-07, UC-10, UC-11)	All	TBD	TBD	TBD	Test
SEC-004	UC-03, UC-05	All	TBD	TBD	TBD	Test
SEC-005	All	All	TBD	TBD	TBD	Inspection
SEC-006	UC-03	Aryaroop	TBD	TBD	TBD	Test
SEC-007	UC-03, UC-05, UC-06	All	TBD	TBD	TBD	Test
SEC-008	UC-07	Bharat / Ayush	TBD	TBD	TBD	Test